



Bachelor Thesis

Quantum network simulator for quantum key distribution

by

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ABSTRACT

Context. Write this at the end

Goal. Write this at the end

Method. Write this at the end

Results. Write this at the end

Conclusions. Write this at the end

1 INTRODUCTION

The secure transmission of sensitive data is one of the key challenges of the modern digital era. Classical cryptographic algorithms such as RSA [6] face a significant threat from the emergence of quantum computers, as their security relies on the assumption that no efficient algorithm exists to invert the encryption using publicly available information. Quantum Key Distribution (QKD) has emerged as a provably secure alternative, grounded in the laws of quantum physics rather than computational complexity.

Many QKD protocols exist [1, 2, 9] that enable the generation of secure keys by exploiting the principles of quantum mechanics. This thesis focuses on implementing the E91 protocol due to its use of entangled qubits and Bell-state verification.

Transitioning these theoretical protocols into scalable real-world networks presents significant engineering challenges. For this reason, quantum network simulators such as SeQUeNCe [12] and QKDNetSim [5] are valuable tools, as they enable controlled testing of new protocols. Nevertheless, the development of such simulators introduces additional challenges, including the accurate modeling of entanglement and distance-dependent fidelity degradation. Due to these limitations and the lack of consensus among existing models, several assumptions were made, which are discussed later in this thesis.

This thesis addresses the following research question: *How can we design and implement an event-driven quantum network simulator capable of evaluating the performance, error-correction overhead, and adversarial resilience of the E91 protocol under realistic hardware imperfections?*

To answer this question, we present the design and implementation of a modular, high-performance quantum network simulator written in Rust [3]. Rust was chosen for its memory safety, high performance, and concurrency support without the overhead of a garbage collector. The core architecture is based on a discrete event-driven simulation engine that uses a priority queue to process events at picosecond resolution (ps). The platform models realistic quantum channels, including distance-dependent entanglement degradation, optical fiber attenuation, and detector cooldown states. It also implements a complete classical post-processing pipeline, including basis reconciliation, CHSH inequality testing, and Cascade error correction. Furthermore, a reactive web interface built

with Leptos and compiled to WebAssembly provides interactive real-time visualization of network topologies.

Through targeted simulation scenarios, we evaluate the performance of the E91 protocol over distances of up to approximately 150 km, assuming a 0.14% fidelity degradation per km based on [7]. To validate the security subsystem, we introduce an active eavesdropper model ("Mallory") performing interception attacks on transmitted qubits. The simulator demonstrates that such interference destroys quantum correlations, reducing the CHSH parameter S below the classical bound ($S < 2$) and triggering protocol termination.

The main contributions of this thesis are as follows:

Event-Driven Simulation Engine: We design and implement a high-performance discrete-event simulation engine in Rust using a priority queue. The engine resolves events at picosecond resolution, accurately modeling asynchronous node behavior, qubit propagation delays, and detector cooldown dynamics.

Realistic Quantum Channel and Device Modeling: We implement detailed abstractions of optical channels and quantum devices, including distance-dependent fidelity degradation, detector cooldown, and Poisson-distributed dark counts.

End-to-End E91 Post-Processing and Security Verification: We construct a complete pipeline for key sifting, CHSH inequality testing, and multi-round Cascade error correction. The simulator reliably detects eavesdropping attempts through degradation of the Bell parameter ($S < 2$).

The remainder of this thesis is organized as follows: Section 2 introduces quantum entanglement, the E91 protocol, and the Cascade error correction algorithm. Section 3 describes the simulator architecture and development challenges. Section 4 presents simulation results and their relevance to real-world scenarios. Finally, Section 7 compares the proposed simulator with existing tools before the concluding section.

2 BACKGROUND

Before discussing the design and evaluation of the simulator, this section introduces the fundamental quantum concepts underlying the E91 key distribution protocol.

In E91, as in any quantum communication system, information is transmitted using **quantum states**. These states encode probability amplitudes whose squared magnitudes determine the likelihood of measurement outcomes. A quantum state that is expressed as a linear combination of basis states is said to be in a **superposition**.

The simplest representation of a quantum system is the **qubit**, a two-level quantum state defined in the computational basis as:

$$|0\rangle, \quad |1\rangle.$$

Using Born's rule, the state of a qubit can be expressed as a linear combination of these basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where $\alpha, \beta \in \mathbb{C}$ and satisfy the probability condition $|\alpha|^2 + |\beta|^2 = 1$.

To operate on qubits, we can use a set of quantum gates analogous to logic gates in classical computing. One example is the **Hadamard (H)** gate, which creates superposition states and is represented by the following matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

For multi-qubit systems, one of the most important gates is the **CNOT** (controlled-NOT) gate, which acts on a two-qubit system. Its matrix representation is given by:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Quantum states can also be represented as vectors in a complex Hilbert space. In this representation, the application of a quantum gate corresponds to matrix multiplication between the gate operator and the state vector.

In order to transform a qubit into a superposition state, where both measurement outcomes have equal probability, the Hadamard gate is applied to a qubit initially prepared in a computational basis state, such as $|0\rangle$.

Representing the basis state $|0\rangle$ as a vector:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

we apply the Hadamard gate via matrix multiplication:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

Entanglement, a key aspect of the E91 protocol, occurs when two qubits are prepared in a quantum state such that their measurement outcomes exhibit correlations, regardless of distance. These entangled two-qubit states are known as Bell states, named after physicist John Bell, who developed Bell's theorem to test the non-classical nature of such correlations.

One of the four maximally entangled two-qubit Bell states is defined as:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

In order to produce a Bell state, you first apply a Hadamard gate, followed by a CNOT gate.

When both qubits of the Bell state are measured in the same basis, the outcomes are perfectly correlated, giving identical results (either 00 or 11) despite each result being individually random. This basis-dependent correlation is the key resource exploited in the E91 protocol, enabling both key generation and eavesdropping detection through later comparison of measurement outcomes.

Building on these quantum mechanical principles, the E91 protocol enables two nodes in the quantum internet, to establish a shared secret key using entangled qubit pairs distributed by a quantum source.

E91 relies on the distribution of entangled states, typically Bell states, such that correlations between measurement outcomes can be used both for key generation and for detecting the presence of an eavesdropper.

In the standard setting, a trusted, usually referred as epr node, distributes entangled qubit pairs to both of the client nodes over a quantum channel. Each party independently measures their qubits using randomly chosen measurement bases. The choice of measurement basis plays a central role in both the generation of the key and the security analysis of the protocol.

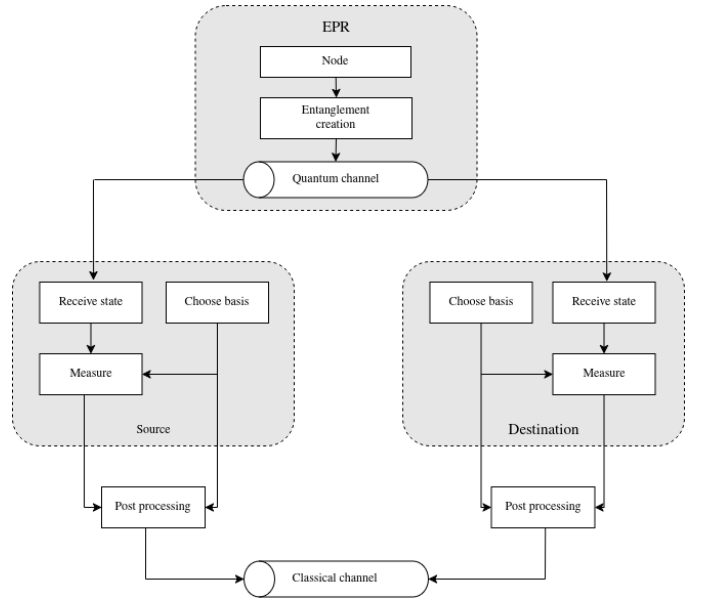


Figure 1: Diagram describing E91 process

Once all entangled pairs have been distributed and measured, both nodes share their chosen measurement bases over a classical channel, which reveals little to no useful information to an eavesdropper.

Using this basis information, the measured qubits can be divided into two groups:

- **Key qubits:** These correspond to measurements where both parties used the same basis. Assuming ideal entanglement and no fidelity loss, the resulting measurement outcomes should be perfectly correlated.
- **Test qubits:** These correspond to measurements performed in different bases. These outcomes are used to evaluate quantum correlations and to detect the presence of an eavesdropper in the communication channel.

In the E91 protocol, the measurement bases correspond to rotations on the Bloch sphere that define the detector settings used to measure incoming qubits. Following [4], the choice of measurement angles that maximizes quantum correlations and leads to a strong violation of the Bell inequality must be chosen in steps of $\frac{\pi}{8}$. The

ones chosen for this simulator were:

$$\begin{aligned}\alpha &= 0 \quad (0^\circ), & \alpha' &= \frac{\pi}{4} \quad (45^\circ), \\ \beta &= -\frac{\pi}{8} \quad (-22.5^\circ), & \beta' &= \frac{\pi}{8} \quad (22.5^\circ),\end{aligned}$$

where α, α' correspond to one party's polarizer settings and β, β' correspond to the other party's polarizer settings.

The CHSH test is an experimental test of Bell's theorem, which shows that no physical theory based on local hidden variables can reproduce all predictions of quantum mechanics.

This result is expressed using the CHSH S -parameter [8], defined as

$$S = |E(A_0, B_0) + E(A_0, B_1)| + |E(A_1, B_0) - E(A_1, B_1)|.$$

Here, $E(A_i, B_j)$ denotes the expectation value of the correlation between measurement settings A_i and B_j .

In any local hidden variable theory, the CHSH inequality imposes the bound

$$S \leq 2,$$

whereas quantum mechanics predicts a stronger limit,

$$S \leq 2\sqrt{2}.$$

Based on [10] when the basis are used as polarizer angles (moving the detector by such an angle) the expectation value can be calculated as:

$$E(a, b) = \cos(2(a - b))$$

Therefore, the CHSH value for our choice of basis is given by

$$\begin{aligned}E(\alpha, \beta) &= \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}, & E(\alpha, \beta') &= \cos\left(-\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}, \\ E(\alpha', \beta) &= \cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}, & E(\alpha', \beta') &= \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}.\end{aligned}$$

Substituting into the CHSH expression,

$$S = |E(\alpha, \beta) + E(\alpha, \beta')| + |E(\alpha', \beta) - E(\alpha', \beta')|,$$

gives

$$S = |\sqrt{2}| + |-\sqrt{2}| = 2\sqrt{2}.$$

If after calculating the S parameter, we get $|S| < 2$, we can assume that the correlation is not maximal and there is a third person in the communication

The Cascade protocol functions as a multi-pass block-parity check designed to reconcile bit errors between Alice and Bob caused by fidelity loss. In the first pass, the sifted bit strings are divided into equal-sized blocks (k_1), and Alice transmits the parity of each block to Bob. For any block where parities disagree—signaling an odd number of errors—the parties execute the BINARY subprotocol. This subprotocol bisects the block and compares the parities of the halves until the exact position of a single erroneous bit is isolated and corrected by Bob, leaving all remaining blocks with an even number of errors.

In subsequent passes ($i > 1$), the block size increases (k_i) and a new mapping function shuffles the bits into different groups. Alice and Bob exchange parities for these new blocks to identify remaining discrepancies. Correcting an error in a later pass alters the parity of the bits in earlier passes, triggering a feedback loop; the protocol identifies the smallest blocks from previous passes that

now have odd errors and recursively runs BINARY to correct them. For this simulator, this framework will run for exactly four rounds of cascade to ensure identical keys before proceeding to hash-based verification.

3 DESIGN

3.1 Modularized Design

To support future adaptations and scalability, the code was structured into layers, each with defined responsibilities and behavior.

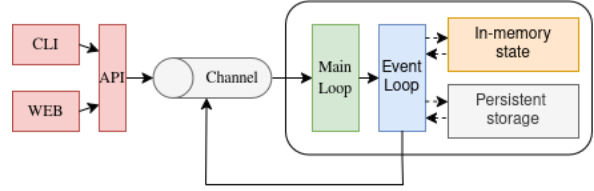


Figure 2: Diagram showing layers of the simulator

1. CLI and Web: These components serve as the primary user interfaces of the simulator. They run on separate threads and provide users with the ability to create nodes, establish links, start simulation runs, and perform many other management tasks.

The CLI operates in a loop that receives commands from the terminal and parses them accordingly. Rust enums provide type safety for commands, while the implementation of the 'From' trait enables efficient conversion between user input and the corresponding command type.

The Web interface runs on an Axum server and serves static HTML content upon request. Once initialized, the JavaScript 'Cytoscape' library enables the visualization and creation of nodes and links within the graph. HTTP is used for standard requests, while WebSockets provide fast communication during simulation runs.

2. API: This layer serves as a bridge between both the CLI/WEB and the channel. It exposes a unified set of functions that translate user requests into events executed in the simulation.

Both interfaces comply with the same function signatures and data models, ensuring consistent behavior without direct coupling between them.

3. Channel: As the simulation engine and the entry points run on different threads, a communication channel must exist between them in order for user requests to be executed by the simulation.

This channel is a single-writer single-reader channel. Instead of sharing state between both layers, which would require locking due to Rust's mutable ownership rules, the entry point threads send event requests that are received by the main loop.

4. Main Loop: This loop is responsible for both receiving requests from the channel and executing the events processed by the event loop.

The loop first checks whether there are any pending events in the channel. If so, the channel is drained until no more events remain. It then checks whether there are any events in the EventLoop; if an event is present, it executes it.

After processing the event, the loop restarts. If no events are present either in the channel or in the event loop, it blocks the

thread while waiting for new events to arrive on the channel. This is intended behavior, as new events can only be triggered by user requests and therefore appear in the channel.

5. Event Loop: This is the event loop responsible for ordering events based on the timestamp at which they must be executed.

It uses a binary heap. A binary heap is a data structure that provides $O(1)$ complexity for accessing the next event and $O(\log n)$ complexity for inserting events in the worst case.

6. In-memory State: This is the state used by the simulation for fast storage and retrieval of information during simulation runs.

It uses multiple HashMaps to store objects such as nodes, links, and measurements. A HashMap provides $O(1)$ complexity for data access and $O(1)$ complexity for data insertion.

As explained later, transitioning from fully persistent storage to an in-memory state resulted in a significant performance improvement.

7. Persistent storage: This is the storage used to persist data that must be shared across simulations. It stores information such as node and link data, generated keys, and other relevant simulation data.

3.2 Design implementations

This section explains how the different objects were simulated.

Nodes and Detectors: Nodes and detectors were implemented as independent objects; however, each node references a detector.

Each node has a `node` type attribute. The two possible types are `Client`, which is responsible for measuring qubits, and `EPR`, which is responsible for generating entangled pairs. A node is locked while it is participating in a simulation run, preventing other simulations from being executed concurrently on the same node.

Each detector has a cooldown rate. Upon reception of a qubit, if the detector is still on cooldown

```
(last_detection_time + cooldown > timestamp)
```

the qubit is dropped.

Dark count rate: The dark count rate is a constant property of the detector. As this rate increases, the average interval between dark count events decreases. Since these intervals should not be fixed but instead occur randomly in time, the process is modeled as a Poisson process. In a Poisson process, the time between consecutive events follows an exponential distribution.

As shown in Figure 3, when the parameter λ becomes smaller (which corresponds to a higher dark count rate), the distribution becomes more concentrated near zero, resulting in shorter intervals between dark count events.

Links: Links model the connection between two nodes. Each link has a distance attribute, which affects both the data propagation delay and the fidelity degradation over the channel.

As explained in Subsection 3.3, a constant fidelity degradation factor of 0.14% per kilometer was used. Therefore, the longer the link, the lower the entanglement fidelity of transmitted qubits.

Entanglement & Measurement: An entangled pair in a Bell state is modeled as a struct instance. This instance stores the measurements from both nodes, and the measurement logic is defined as follows:

- (1) Check whether the current measurement is the first.

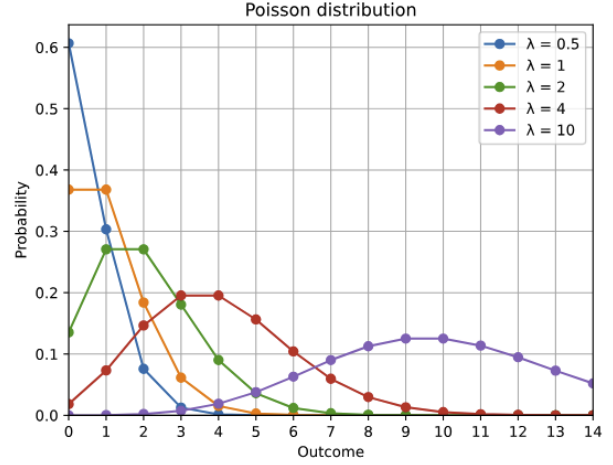


Figure 3: Diagram showing the poisson distribution

If it is the first measurement:

- (a) Choose a random basis.
- (b) Generate a random number between 0 and 1.
- (c) Store the measurement in the corresponding measurement attribute of the pair.

If it is the second measurement:

- (a) Calculate the updated fidelity after distance-based degradation.
- (b) Choose a random basis.
- (c) Compute the basis difference between the first measurement basis and the current one.
- (d) Compute the entanglement probability.
- (e) Based on this probability, either copy the first measurement outcome or take the opposite result.
- (f) Store the measurement in the corresponding measurement attribute of the pair.

The entanglement probability function is as follows:

$$P_{\text{same}} = F \cdot \cos^2(\alpha - \beta) + (1 - F) \cdot 0.5$$

Where:

- α and β are the chosen angles.
- F is the fidelity after degradation.
- $\cos(\cdot)$ represents the correlation function.

This function behaves correctly because when the fidelity is zero, the probability becomes 0.5, meaning the output is completely random.

Mallory: The most appropriate way to model the presence of Mallory in a connection was to define links as either secure or insecure, and to execute different logic depending on the type of link.

When a link is defined as insecure, a measurement is performed and stored in the pair before the actual measurement. Then, in subsequent measurements, the basis calculation and entanglement probability are computed using Mallory's measurements instead of the actual client node measurements.

This result in a completely random outcome.

3.3 Obstacles

This section exposes some of the many obstacles found during the design and development process.

The first obstacle arose during the design phase. During the investigation, it was found that many simulators model fidelity degradation through multiple physical effects, such as fiber optic attenuation, quantum memory efficiency, and other channel imperfections. However, since this project focuses on the E91 protocol simulation and CHSH inequality validation, a highly detailed physical model of fidelity degradation was not required.

After extensive research, the paper [7] showed that after 100 km of fiber optic transmission, the entanglement fidelity decreases to approximately 86%, corresponding to a reduction of about 0.14% per kilometer. Therefore, a constant fidelity degradation rate of 0.14% per kilometer was chosen for the simulation.

During the development of the simulator, several obstacles were encountered, mainly due to Rust's programming rules.

In a simulation where events are scheduled based on runtime conditions, the system needs a way to execute functions dynamically. In Python, this can be achieved using reflection, where a dotted string path is resolved at runtime into a callable object. However, Rust is a statically compiled language and does not provide built-in runtime reflection of function names.

The solution was to create a registry that stored a function HashMap. This HashMap would map a EventName instance to its corresponding function stored in a Box pointer.

The last obstacle was performance. At the beginning, persistent storage was used as the storage for everything. This meant that not only nodes and links were stored, but also measurements and entangled pairs. As a result, a simulation with 10,000 qubits took more than 5 minutes to complete due to database accesess and locks between threads.

The solution was to move away from a shared database state between threads and instead adopt an in-memory HashMap-based state managed directly by the simulation loop.

4 EVALUATION

Discuss the design of your experiments, the results you obtained, and how they help in evaluating the claims you made in the introduction. You may also use the evaluation results in this section to justify your design choices or assess the contributions of different aspects of your design towards the overall goals.

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5 DISCUSSION

Here you put your results in context (possibly grouped by research question). Usually, this section focuses on analyzing the implications of the proposed work for current and future research and for practitioners.

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6 THREATS TO VALIDITY

Report about each type of threat to the validity of the experiment, according to the classification framework proposed by Wohlin *et al.* [11].

6.1 Internal Validity

6.2 External Validity

6.3 Construct Validity

6.4 Conclusion Validity

7 RELATED WORK

Describe here scientific papers similar to your experiment, both in terms of goal and methodology. One paragraph for each paper (we expect about 5-8 papers to be discussed). Each paragraph contains: (i) a brief description of the related paper and (ii) a black-on-white description about how your work differs from the related paper. You may place this section immediately after the Background section, if necessary.

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8 CONCLUSION

Briefly summarize your contributions, and share a glimpse of the implications of this work for future research.

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